EECS492 Final

Planning

~agent: goal-based
knowledge-based
~task: { init/goal state
action

⇒ find a course of action

Classical Planning

fully observable, finite deterministic, static, discrete, single~, known

*state: conjunctions of function free ground positive literals with closed world semantics. (no ¬)

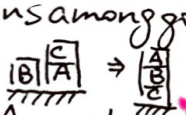
*goal: conjunctions of literals or negations

Planning Domain Def Lan

action { name
precondition
effect

State space search

{ progression (heuristic)
regression (only consider relevant action; deal with interactions among goal states)



Sussman Anomaly

Graph Plan

mutex Action:
Inconsistent effects
(produce complementary propositions)
Interference (one negates the precondition of other)
Competing Needs (complementry preconditions)

Proposition: Compliments
Inconsistent Support

All pairs of actions that produce the pair are mutex

*nogood: impossible sub-goal combinations.

Propositional Planning: SAT

int state ∧ goal* ∧ domain axiom

successor state ~, pre-condition ~, action exclusion

Partial Order Planning

causal link

solution { Complete } exist
consistent: no contradiction/constraint

threat (衝突) { promotion
demotion

Planning Under Uncertainty

sensorless/conformant ~

conditional/contingent ~
*percept

replanning
{ action monitoring
plan monitoring

Probability

Conditional ~ $Pr(A|B) = \frac{Pr(A,B)}{Pr(B)}$

Bayes' Theorem

$P(A|B) = P(A,B)/P(B)$

$= P(B|A)P(A)/P(B)$ $\sum_a P(B|A=a)$

Bayesian Networks

Independence: $P(A|B) = P(A)$

$\Rightarrow P(A,B) = P(A)P(B)$

Conditional ~: $P(A|B,C) = P(A|C)$

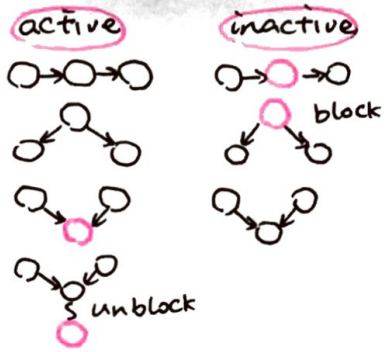
Applying chain rule:

$P(x_1,\dots,x_n) = P(x_1|x_2,\dots,x_n)\cdot P(x_2|x_3,\dots,x_n)\cdot P(x_3|x_4,\dots,x_n)\cdots$

d-seprated ⇒ conditional independence

$IS(a,S,b)$ iff no active path between each tuple is active

iff $\forall$ undirected path blocked (all) no path unblocked.



Law of Total Pr

$P(W) = \sum_r \sum_s \sum_c P(W,r,s,c)$

(can move summation inward)

Known Value Iteration

$U(i) = R(i) + \gamma \max_a \sum_k T_{ik}(a) U(k)$

transition probability given choice action

Bellman Equation

Variable Clustering

Bayesian Network

⇒ polytree

Markov Decision Process

$U'(s) = R(s) + \gamma \max_a \sum P(s'|s,a) U(s')$

Value of Perfect Info

$\sum_e Pr(E'=e|E) EU(\alpha_e|E,e) - EU(\alpha,E)$

Uncertainty/Entropy

$H(X) = -\sum P(x_i)\log_2 P(x_i)$

$Gain(S|A) = H(S) - H(S|A)$

Learning

Decision Tree

Entropy $I(q) = -q\log_2 q - (1-q)\log_2(1-q)$

Before: $I(p/(p+n))$

After: $\frac{p_t+n_t}{p+n} I\left(\frac{p_t}{p_t+n_t}\right) + \frac{p_f+n_f}{p+n} I\left(\frac{p_f}{p_f+n_f}\right)$

Ensemble Learning

AdaBoost: great for weak learning algorithm; (large number of iteration

For each iteration k,

Find $H_k(x)$ with $W_k(x)$

$\varepsilon_k = \sum W(\text{incorrect})$

$\alpha_k = \ln(1-\varepsilon_k/\varepsilon_k)$

update correct training instance

$W_{k+1} = W_k \times \varepsilon_k/(1-\varepsilon_k)$

*re-normalize

$H_{FINAL}(x) = \text{sign}(\sum \alpha_i H_i(x))$

Reinforcement Learning

(Q-Learning) (Adaptive Dynamic Programming learns model)

*model free

*no reward/transition func

$Q = q + \alpha\cdot(R + (\gamma\, q') - q)$

for state and action — best q'

{ exploration
exploitation

Probably Approximately Correct Learning

n attributes (input) that take k value each.

output one of n class es

⇒ m^{k^n} possible hypothesis

Given x examples, still have m^{k^n}/m^x hypothesis that are consistent.

Approximately correct.

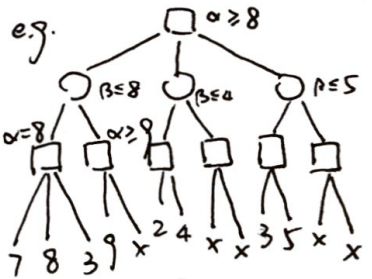
Naive Bayes

greatest posterior probability (joint probability of the example given that class, and of observing each attribute value)

$\hat{y} = \text{argmax}\, p(C_k)\prod p(x_i|C_k)$

Alpha-Beta Pruning

adversarial MINIMAX

$O(b^m) \Rightarrow O(b^{m/2}) \sim O(b^{3m/4})$

e.g.